## 4-4b Three Steps to Analyzing Changes in Equilibrium

So far, we have seen how supply and demand together determine a market's equilibrium, which in turn determines the price and quantity of the good that buyers purchase and sellers produce. The equilibrium price and quantity depend on the positions of the supply and demand curves. When some event shifts one of these curves, the equilibrium in the market changes, resulting in a new price and a new quantity exchanged between buyers and sellers.



When analyzing how some event affects the equilibrium in a market, we proceed in three steps. First, we decide whether the event shifts the supply curve, the demand curve, or, in some cases, both. Second, we decide whether the curve shifts to the right or to the left. Third, we use the supply-and-demand diagram to compare the initial equilibrium with the new one, which shows how the shift affects the equilibrium price and quantity. Table 3 summarizes these three steps. To see how this recipe is used, let's consider various events that might affect the market for ice cream.

Table 3

**Three Steps for Analyzing Changes in Equilibrium**

1. Decide whether the event shifts the supply or demand curve (or perhaps both).

2. Decide in which direction the curve shifts.

3. Use the supply-and-demand diagram to see how the shift changes the equilibrium price and quantity.

**Example: A Change in Market Equilibrium Due to a Shift in Demand**

Suppose that one summer the weather is very hot. How does this event affect the market for ice cream? To answer this question, let's follow our three steps.

1. The hot weather affects the demand curve by changing people's taste for ice cream. That is, the weather changes the amount of ice cream that people want to buy at any given price. The supply curve is unchanged because the weather does not directly affect the firms that sell ice cream.

2. Because hot weather makes people want to eat more ice cream, the demand curve shifts to the right. Figure 10 shows this increase in demand as a shift in the demand curve from $D_1$ to $D_2$. This shift indicates that the quantity of ice cream demanded is higher at every price.

   Figure 10

   **How an Increase in Demand Affects the Equilibrium**

   An event that raises quantity demanded at any given price shifts the demand curve to the right. The equilibrium price and the equilibrium quantity both rise. Here an abnormally hot summer causes buyers to demand more ice cream. The demand curve shifts from $D_1$ to $D_2$, which causes the equilibrium price to rise from $\$4$ to $\$5$ and the equilibrium quantity to rise from $7$ to $10$ cones.

3. At the old price of $\$4$, there is now an excess demand for ice cream, and this shortage induces firms to raise the price. As Figure 10 shows, the increase in demand raises the equilibrium price from $\$4$ to $\$5$ and the equilibrium quantity from $7$ to $10$ cones. In other words, the hot weather increases both the price of ice cream and the quantity of ice cream sold.

**Shifts in Curves versus Movements along Curves**

Notice that when hot weather increases the demand for ice cream and drives up the price, the quantity of ice cream that firms supply rises, even though the supply curve remains the same. In this case, economists say there has been an increase in "quantity supplied" but no change in "supply."

*Supply* refers to the position of the supply curve, whereas the *quantity supplied* refers to the amount producers wish to sell. In this example, supply does not change because the weather does not alter firms' desire to sell at any given price. Instead, the hot weather alters consumers' desire to buy at any given price and thereby shifts the demand curve to the right. The increase in demand causes the equilibrium price to rise. When the price rises, the quantity supplied rises. This increase in quantity supplied is represented by the movement along the supply curve.

To summarize, a shift *in* the supply curve is called a "change in supply," and a shift *in* the demand curve is called a "change in demand." A movement *along* a fixed supply curve is called a "change in the quantity supplied," and a movement *along* a fixed demand curve is called a "change in the quantity demanded."

**Example: A Change in Market Equilibrium Due to a Shift in Supply**

Suppose that during another summer, a hurricane destroys part of the sugarcane crop and drives up the price of sugar. How does this event affect the market for ice cream? Once again, to answer this question, we follow our three steps.

1. The change in the price of sugar, an input for making ice cream, affects the supply curve. By raising the costs of production, it reduces the amount of ice cream that firms produce and sell at any given price. The demand curve does not change because the higher cost of inputs does not directly affect the amount of ice cream consumers wish to buy.

2. The supply curve shifts to the left because, at every price, the total amount that firms are willing and able to sell is reduced. Figure 11 illustrates this decrease in supply as a shift in the supply curve from $S_1$ to $S_2$.

   **Figure 11**

   **How a Decrease in Supply Affects the Equilibrium**

An event that reduces quantity supplied at any given price shifts the supply curve to the left. The equilibrium price rises, and the equilibrium quantity falls. Here an increase in the price of sugar (an input) causes sellers to supply less ice cream. The supply curve shifts from $S_1$ to $S_2$, which causes the equilibrium price of ice cream to rise from $\$4$ to $\$5$ and the equilibrium quantity to fall from $7$ to $4$ cones.

3. At the old price of $\$4$, there is now an excess demand for ice cream, and this shortage causes firms to raise the price. As Figure 11 shows, the shift in the supply curve raises the equilibrium price from $\$4$ to $\$5$ and lowers the equilibrium quantity from $7$ to $4$ cones. As a result of the sugar price increase, the price of ice cream rises, and the quantity of ice cream sold falls.

**Example: Shifts in Both Supply and Demand**

Now suppose that the heat wave and the hurricane occur during the same summer. To analyze this combination of events, we again follow our three steps.

1. We determine that both curves must shift. The hot weather affects the demand curve because it alters the amount of ice cream that consumers want to buy at any given price. At the same time, when the hurricane drives up sugar prices, it alters the supply curve for ice cream because it changes the amount of ice cream that firms want to sell at any given price.

2. The curves shift in the same directions as they did in our previous analysis: The demand curve shifts to the right, and the supply curve shifts to the left. Figure 12 illustrates these shifts.

   **Figure 12**

   **A Shift in Both Supply and Demand**

   Here we observe a simultaneous increase in demand and decrease in supply. Two outcomes are possible. In panel (a), the equilibrium price rises from $P_1$ to $P_2$, and the equilibrium quantity rises from $Q_1$ to $Q_2$. In panel (b), the equilibrium price again rises from $P_1$ to $P_2$, but the equilibrium quantity falls from $Q_1$ to $Q_2$.

3. As Figure 12 shows, two possible outcomes might result depending on the relative size of the demand and supply shifts. In both cases, the equilibrium price rises. In panel (a), where demand increases substantially while supply falls just a little, the equilibrium quantity also rises. By contrast, in panel (b), where supply falls substantially while demand rises just a little, the equilibrium quantity falls. Thus, these events certainly raise the price of ice cream, but their impact on the amount of ice cream sold is ambiguous (that is, it could go either way).

**Summary**

We have just seen three examples of how to use supply and demand curves to analyze a change in equilibrium. Whenever an event shifts the supply curve, the demand curve, or perhaps both curves, you can use these tools to predict how the event will alter the price and quantity sold in equilibrium. Table 4 shows the predicted outcome for any combination of shifts in the two curves. To make sure you understand how to use the tools of supply and demand, pick a few entries in this table and make sure you can explain to yourself why the table contains the prediction that it does.

Table 4

**What Happens to Price and Quantity When Supply or Demand Shifts?**

As a quick quiz, make sure you can explain at least a few of the entries in this table using a supply-and-demand diagram.

| | **No Change in Supply** | **An Increase in Supply** | **A Decrease in Supply** |
|---|---|---|---|
| No Change in Demand | *P* same<br>*Q* same | *P* down<br>*Q* up | *P* up<br>*Q* down |
| An Increase in Demand | *P* up<br>*Q* up | *P* ambiguous<br>*Q* up | *P* up<br>*Q* ambiguous |
| A Decrease in Demand | *P* down<br>*Q* down | *P* down<br>*Q* ambiguous | *P* ambiguous<br>*Q* down |

QuickQuiz

10. The discovery of a large new reserve of crude oil will shift the ____ curve for gasoline, leading to a ____ equilibrium price.

    a. supply; higher

    b. supply; lower

    c. demand; higher

    d. demand; lower

11. If the economy goes into a recession and incomes fall, what happens in the markets for inferior goods?

    a. Prices and quantities both rise.

    b. Prices and quantities both fall.

    c. Prices rise and quantities fall.

d. Prices fall and quantities rise.

12. Which of the following might lead to an increase in the equilibrium price of jelly and a decrease in the equilibrium quantity of jelly sold?

    a. an increase in the price of peanut butter, a complement to jelly

    b. an increase in the price of Marshmallow Fluff, a substitute for jelly

    c. an increase in the price of grapes, an input into jelly

    d. an increase in consumers’ incomes, as long as jelly is a normal good

13. An increase in ____ will cause a movement along a given supply curve, which is called a change in ____ .

    a. supply; demand

    b. supply; quantity demanded

    c. demand; supply

    d. demand; quantity supplied